

Mikhail Razakov

Compiler & Program Analysis Engineer

razakov.mikhail@outlook.com
t.me/Micodiy
github.com/Misha1302

Compiler and program-analysis engineer: LLVM 22 optimization at MCST and C# static analysis at ISP RAS/SharpChecker. Public projects demonstrate conservative legality, Roslyn fixed-point data flow, and x86-64 register allocation; beyond compiler work I own a stateful backend with idempotency, recovery, and rollback.

EXPERIENCE

- 2026 — present** **ISP RAS — Static Analysis Engineer**
- Contribute to SharpChecker, ISP RAS's industrial Roslyn-based static-analysis platform for C#.NET.
 - Current tasks include symbolic-computation / symbolic-execution work in C# program analysis.
- July — August 2026** **MCST — Compiler Engineering Intern**
- Implemented an LLVM 22 LICM pass with loop analysis, side-effect/speculative-safety checks, and hoisting only proven invariants to preheaders.
 - Developed a conservative interprocedural global-IV prototype with APInt affine evolution, transitive call effects, a separate legality phase, and LLVM IR transformation; unsupported cases are rejected unchanged.
- 2026 — present** **Independent Engineering — stateful systems & reliability**
- Own a stateful subscription backend: explicit payment/access states, idempotent reconciliation, migrations, backup/restore, health gates, and rollback; release gate: 388 Python + 148 .NET tests.

PROJECTS

LLVM Interprocedural Global IV Optimization

An LLVM 22 pass localizes supported induction-like globals after interprocedural effects/affine analysis and a separate legality decision; APInt models fixed-width wraparound.

29 positive/negative regression cases: global-iv,verify, structural expectations, idempotence, and observable before/after execution comparison.

DerefAfterNullAnalyzer

A Roslyn ControlFlowGraph analyzer propagates null state over conditional edges to a fixed point, handles loops/back edges, and merges states from multiple predecessors.

Handles method/property/index/event dereferences and invalidation after assignment/ref/out; dedicated analyzer tests use Microsoft.CodeAnalysis.CSharp.Testing.

x86-64 Codegen & Register Allocation Lab

A compiler backend with SSA/CFG validation, liveness/interference analysis, linear-scan and seeded simulated-annealing allocators, spills, phi lowering, and a SysV x86-64 emitter.

An independent assignment verifier plus native-vs-interpreter differential execution; tests cover branches, loops, spills, and parallel phi moves.

SKILLS

Compiler / IR

C++23, LLVM 22/IR, APInt, interprocedural analysis, CFG/SSA, dominators, loops, LICM, legality, LLVM Verifier

Program Analysis

C#.NET, Roslyn CFG, fixed-point data flow, symbolic execution, conditional edges, loops/back edges, state propagation/joins

Backend / Codegen

Rust, SysV x86-64, liveness/interference, linear scan, simulated annealing, spills, phi lowering, differential execution

Infrastructure / Reliability

typed contracts, deterministic planning, interpreter/CIL parity, state machines, idempotency, recovery, migrations, rollback

EDUCATION

HSE University — Software Engineering, 2026–2030

RECOGNITION

LangDev'26 — Build the Language, Then Make the Abstractions Disappear

Accepted talk on UniversalToolchain's architecture; LangDev'26 takes place October 8–9, 2026 in Málaga.

UniversalToolchain — Baltic Science and Engineering Competition

First Degree Diploma and Grand Prize "Perfection as Hope" for UniversalToolchain.

MEPhI Junior

Overall winner; First Degree Diploma, 96/100 for UniversalToolchain.

Moscow / remote